Technical Master Compliance & Safety Report: ECU (FANS_DYN10)

Parameter	Specification
Vehicle Subsystem	Electronic Control Unit (ECU - FANS_DYN10)
Production Target	ECU_FW (ESP-IDF v5 + FreeRTOS SMP)
Applied Standards	MISRA-C:2012, ISO 26262 (ASIL-B Principles), Zero Dynamic Memory, 100 Hz Deterministic FreeRTOS
CI/CD Build & Test Status	**PASSED** (Compilation with `-Werror`, static analysis, and 100% host unit tests passing)

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1. Executive Summary & System Architecture

The **ECU (FANS_DYN10)** is an automotive embedded electronic control unit responsible for:

1. **High-Precision Thermal Acquisition**: Interfacing with the powertrain temperature sensors (Motor and Inverter) via the industrial 16-bit SPI **ADS8688** ADC and Bosch NTC thermistors.

2. **Closed-Loop Active Thermal Management (PID)**: Driving dual LEDC hardware PWM channels at 50 Hz with 14-bit resolution to command Electronic Speed Controllers (ESCs) for liquid/air cooling.

3. **High-Speed Suspension Strain Gauge Acquisition (100 Hz)**: Synchronously sampling 4 Suspension Travel Sensors (STS: Rear-Right, Rear-Left, Front-Right, Front-Left) for vehicle dynamics analysis.

4. **Deterministic CAN / TWAI Bus Networking (500 kbps)**: Broadcasting real-time telemetry (IDs `0x401` and `0x402`), handling diagnostic trouble codes (DTCs) on ID `0x503`, and receiving vehicle state frames (`0x021`) from the Motor Control Unit (MCU).

5. **Safety OTA Interlocking**: Hardware-level and software-level locking preventing remote firmware updates while the vehicle is in `READY_TO_DRIVE` state.

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2. Exhaustive Fault Management Matrix (DTC & Safe States)

The ECU firmware implements an isolated, centralized fault manager (`fault_manager.c`) that classifies every anomaly, activates deterministic hardware failsafes, and broadcasts structured Diagnostic Trouble Codes (DTC) over the CAN bus.

DTC Code	Fault Identifier	Category	Priority	Exact Trigger Condition	System & Physical Hardware Reaction	Recovery / Reset Mechanism	Broadcasted CAN Frame

`201`	`FAULT_CODE_MOTOR_NTC_FAIL`	`FAULT_CAT_HARDWARE`	`HIGH`	≥ 3 consecutive failed reads on Motor NTC channel (ADS8688 CH0 voltage outside [0.01V, 5.11V] or computed temperature outside [-40^°C, +130^°C]).	Activates Thermal Failsafe State: Immediately commands Motor cooling fan to 10% duty and ramps up by +10% per second until reaching **100% continuous maximum cooling power**. Prevents motor thermal runaway in case of broken wire or loose sensor.	Automatic recovery when both Motor and Inverter NTC sensors provide valid temperatures simultaneously for 1 full second.	CAN ID `0x503`
							 `Byte 0 = 1` (Failsafe Active)
							 `Byte 1 = 0` (HW)
							 `Byte 2 = 1` (High Prio)
							 `Bytes 3-4 = 201`
							 `Byte 5 = FanMotor%`
							 `Byte 6 = FanInv%`

`202`	`FAULT_CODE_INV_NTC_FAIL`	`FAULT_CAT_HARDWARE`	`HIGH`	≥ 3 consecutive failed reads on Inverter NTC channel (ADS8688 CH7).	Activates Thermal Failsafe State: Ramps Inverter cooling fan to **100% maximum power** (+10%/s slew). Protects power semiconductors (IGBT/SiC modules) from unmonitored overheating.	Automatic recovery when both sensors return within valid calibration bounds.	CAN ID `0x503`
							 `Byte 0 = 1`
							 `Bytes 3-4 = 202`

`203`	`FAULT_CODE_ADS8688_SPI_ERR`	`FAULT_CAT_HARDWARE`	`HIGH`	SPI peripheral communication timeout or invalid NOP response from ADS8688 ADC chip.	Invalidates all temperature and suspension strain gauge telemetry. Forces both cooling fans to **100% continuous maximum duty**.	Requires SPI peripheral re-initialization or hard hardware reboot.	CAN ID `0x503`
							 `Byte 0 = 1`
							 `Bytes 3-4 = 203`

`204`	`FAULT_CODE_TWAI_BUS_OFF`	`FAULT_CAT_COMMUNICATION`	`HIGH`	TWAI/CAN peripheral enters `TWAI_STATE_BUS_OFF` due to physical bus degradation, missing termination resistors, or excessive frame collisions.	Invokes `twai_initiate_recovery()`, resets internal transmission buffers, and restarts CAN interface automatically.	Automatic once CAN controller completes the 128 occurrences of 11 consecutive recessive bits required by ISO 11898-1.	CAN ID `0x503` immediately upon bus recovery

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3. CAN / TWAI Bus Network Architecture (500 kbps)

All frames adhere to standard 11-bit CAN identifiers and are serialized with strict endianness and static memory buffers.

3.1. Transmitted CAN Frames

CAN ID	Message Name	DLC	Rate	Target Nodes	Payload Layout (Byte-by-Byte)	Resolution & Units
`0x401`	`ECU_TEMPS`	4	1 Hz	MCU, Dashboard, Data Logger, Telemetry	`Byte 0-1` : Motor Temperature (Big-Endian int16) `Byte 2-3` : Inverter Temperature (Big-Endian int16)	1\ LSB = 1^°C Range: -40^°C \dots +130^°C
`0x402`	`ECU_STS_GAUGES`	8	100 Hz	Telemetry, Vehicle Dynamics Logger	`Byte 0-1` : STS Rear-Right (Big-Endian uint16) `Byte 2-3` : STS Rear-Left (Big-Endian uint16) `Byte 4-5` : STS Front-Right (Big-Endian uint16) `Byte 6-7` : STS Front-Left (Big-Endian uint16)	1\ LSB = 1\ ADC Count Range: 0 \dots 65535 (16-bit raw)
`0x503`	`ECU_DIAGNOSTIC_DTC`	8	10 Hz / On-Fault	Safety Master, Dashboard, Telemetry	`Byte 0` : Failsafe Active (1 = Active, 0 = Normal) `Byte 1` : Category (0=HW, 1=Comm, 2=Res, 3=Timing) `Byte 2` : Priority (0=Low, 1=High) `Byte 3-4` : Active DTC Code (Big-Endian uint16) `Byte 5` : Motor Fan Duty (0 \dots 100\%)	`Byte 6` : Inverter Fan Duty (0 \dots 100\%) `Byte 7` : Cumulative Fault Count

3.2. Received CAN Frames

CAN ID	Sender Node	Key Payload Fields	ECU Action & Functional Behavior
`0x021`	**MCU** (Motor Control Unit)	`Byte 6` : Vehicle State & R2D (4 = READY_TO_DRIVE).	Interlocks the Over-The-Air (OTA) firmware update service (`ota_service.c`). Flashing is permanently rejected whenever the vehicle is operating or in ready-to-drive mode.

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4. Unit Test Validation Suite (Unity Test Framework)

All mathematical models, signal processing algorithms, and safety failsafes are validated in PlatformIO native x86 simulation using the **Unity** testing framework:

```
Processing * in native environment
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Building...
Testing...
test/test_main.cpp:88: test_ads8688_bosch_ntc_conversion      [PASSED]
test/test_main.cpp:89: test_fan_driver_esc_scaling            [PASSED]
test/test_main.cpp:90: test_fan_driver_slew_rate              [PASSED]
test/test_main.cpp:91: test_pid_cooling_reverse_action        [PASSED]
test/test_main.cpp:92: test_fault_manager_failsafe_escalation [PASSED]
----------------------- 5 Tests 0 Failures 0 Ignored ---------------------------
OK
```

Detailed Test Case Specifications

1. **`test_ads8688_bosch_ntc_conversion`** :

- Evaluates Steinhart-Hart / Bosch logarithmic interpolation table across calibrated operating points: verifies 20^°C at 2500\ Ω and 50^°C at 834\ Ω.
- Validates that out-of-range or negative resistance values reliably produce `NAN` to trigger fault handling.

2. **`test_fan_driver_esc_scaling`** :

- Asserts pulse-width output: 0\% \to 1000\ µs, 50\% \to 1570\ µs, 100\% \to 2000\ µs.
- Verifies 14-bit timer duty calculation at 50 Hz (T = 20000\ µs), producing exact 819 integer duty counts for minimum 1000\ µs arming pulse.

3. **`test_fan_driver_slew_rate`** :

- Checks rate-limiter slew dynamics (20\%/s): verifies that a 0 \to 100\% target step is clamped to ≤ 2.0\% variation over a 100 ms interval.

4. **`test_pid_cooling_reverse_action`** :

- Verifies reverse-acting PID control logic with a 45^°C setpoint: outputs 0.0\% duty below setpoint (30^°C) and delivers positive proportional-integral cooling duty when above setpoint (55^°C).

5. **`test_fault_manager_failsafe_escalation`** :

- Verifies fault reporting, priority classification, failsafe flag latching, and DTC diagnostic record formatting.

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5. Industrial Standards & Safety Compliance Certification

The ECU production codebase complies with standard automotive embedded requirements:

1. **Zero Dynamic Memory Allocation**:

- No usage of `malloc` , `calloc` , `realloc` , `free` , `new` , or `delete` .
- All FreeRTOS primitives (Tasks, Queues, Semaphores) instantiated via `xTaskCreateStaticPinnedToCore` , `xQueueCreateStatic` , and `xSemaphoreCreateMutexStatic` .

2. **Temporal Determinism & Multi-Core Execution**:

- Core 0 runs deterministic 100 Hz analogue acquisition and 1 Hz PID thermal control with `vTaskDelayUntil` .
- Core 1 handles asynchronous CAN transmission/reception with dedicated static queues (`ipc.c`), preventing latency spikes.

3. **MISRA-C:2012 Compliance**:

- Strict explicit typing (`uint8_t` , `int16_t` , `uint32_t` , `float` , `double`).
- All switch statements provide default handlers; no uninitialized variables or implicit sign conversions.

4. **Zero Compiler Warnings (`-Werror`)**:

- Fully compliant build with `-Wall -Wextra -Werror -Wmissing-field-initializers` in ESP-IDF v5 and Native GCC.